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Roll No

EE-6001-CBGS

B.E. VI Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Electronic Magnetic Field Theory

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Describe cylindrical coordinate system and also deduce its conversion into spherical coordinate system. 7
b) State and explain Divergence theorem and Stoke's theorem. Give the relevance of these theorems to electromagnetic problems. 7
2. a) State and explain Coulomb's law. 7
b) Point charge $1\mu\text{C}$ and $2\mu\text{C}$ are located at $(3, 2, -1)$ and $(-1, -1, 4)$ respectively. Calculate the electric force on at 10 nC charge located at $(0, 3, 1)$ and the electric field intensity at that point. 7
3. a) Derive Laplace's and Poisson's equation and show that they have unique solution. 7
b) State and explain Gauss's flux theorem. Also write limitations of Gauss's law. 7
4. a) Find the store energy in a system of four identical charges $Q = 2\text{ nC}$ at the corners 1m on a side. 7
b) Explain the concept of energy density for a parallel plate capacitor. 7

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5. a) Explain the method of electrical images and discuss its applications in the study of electromagnetic problems. 7
b) Explain boundary relation in magnetic fields and hence explain magnetic boundary conditions. 7
6. a) A current element $I\Delta l = 2\pi(0.6\hat{i}_x - 0.8\hat{i}_y)\mu\text{A}$ is situated at a point (4, -2, 3). Find the incremental field ΔH at a point (1, 3, 2). 7
b) State and explain Ampere's law both in integral and differential form as used in magnetic fields. 7
7. a) Deduce Maxwell's equation in differential and integral form. 7
b) Determine the inductance of solenoid, toroid and coaxial cable with equations. 7
8. a) State and prove pointing vector theorem. 7
b) Explain the concept of polarization of waves. 7
